HARTS Engineering Report for Human-to-Robot Container Handover

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***Abstract*—This short report describes HARTS, a hand-referenced reliability-gated binocular target-stream estimator and integrated RGB-D-only handover stack for human-to-robot container handover. The stack uses a UR5 robot, two eye-to-hand Intel RealSense D435i RGB-D cameras, a compliant CAN-controlled parallel gripper, ROS 2 Humble, semantic cup segmentation, MediaPipe hand landmarks, and bounded RTDE `servoL` control. The core contribution is not only object detection, but a control-ready 3D target stream that remains fresh and bounded during hand-object occlusion. In replay logs the final stream reached 92.89% valid output at 96.52 Hz, with 7.74 mm RMS residual and only 10 target jumps larger than 20 mm. In full-system validation the deployed stack submitted 288 standardized configurations, achieved a task score of 95.15, and completed 21 of 24 live valid handovers.**

***Index Terms*—human-to-robot handover, RGB-D perception, target stream, visual servoing, compliant gripper**

I. System Overview

HARTS denotes the hand-referenced reliability-gated binocular target-streaming system used in this report. It combines semantic container evidence, stereo fingertip geometry, per-view reliability assessment, freshness gating, workspace constraints, anchor-bounded publication, and causal temporal filtering to produce the servo target consumed by the robot controller.

The task is to receive a naturally handed-over container without motion capture, external markers, wrist force-torque sensing, or tactile arrays. The robot therefore needs a continuous and current 3D target for visual servoing. We define the perception output as a target stream

$$\mathscr{S} = \{(t_k, \mathbf{p}_k, v_k, q_k)\}_{k=1}^{N}, \tag{1}$$

where $\mathbf{p}_k$ is the robot-base target position, v_k is the validity flag, and q_k records freshness and quality. The engineering goal is to keep this stream available, fresh, smooth, and physically reachable while the human hand partially occludes the object.

The runtime is organized as a ROS 2 system. The perception node reads both RealSense devices, projects RGB-D observations into the robot base frame, and publishes cup, hand, and diagnostic points. Bridge nodes convert the selected point to `/robot_wrist_pos` and `/external_target_pose`. A staged scheduler sends bounded Cartesian targets to the UR5 and synchronized opening/closing commands to the gripper bridge.

The implementation is intentionally modular so that perception, target selection, arm motion, and gripper motion can be replayed or disabled independently. Perception publishes both the final target and single-eye diagnostic streams, which made it possible to test fusion failures without changing robot code. The robot side consumes only the final target and its freshness state, so detector replacement or parameter tuning does not change the control interface.

Fig. 1. Compliant parallel gripper used for receiving containers. The large contact pads and mechanical compliance tolerate residual visual-servo error; motor opening, speed, and power provide the deployed contact proxy.

II. Perception and Trained Models

The perception stack combines trained visual models and deterministic control-safety filters. Semantic segmentation is performed with YOLO-based models. The deployed open-vocabulary model is `yoloe-26s-seg.pt`; the experiment tree also records task-specific cup segmentation weights for four-cup empty/rice and empty-only settings. MediaPipe Hands supplies hand landmarks; no learned tactile model is used.

TABLE I
Hardware, workstation, and setup

Item	Deployed configuration
Robot	UR5, 6 DoF, tabletop handover setup; Cartesian receiving and delivery through RTDE `servoL`.
Sensors	Two Intel RealSense D435i RGB-D cameras, serials 335222072264 and 346222073199, eye-to-hand mounting, 640×480 RGB/depth at 30 Hz with depth aligned to color.
Calibration	Per-camera `T_base_camera`; Daniilidis hand-eye calibration. Mean translation errors: 2.34 mm and 2.30 mm; mean rotation errors: 0.30° and 0.45°.
End effector	Thumanoid CPG-F2 compliant two-finger parallel gripper, flange-mounted to UR5, single CAN motor controller, 0–100 mm commanded opening.
Workstation	ASUS ROG Strix SCAR 18 G835LX, Intel Core Ultra 9 275HX, 24 Linux CPU threads, 62 GiB RAM, Ubuntu 22.04.5 LTS, ROS 2 Humble.
GPU	NVIDIA GeForce RTX 5090 Laptop GPU (24 463 MiB VRAM), NVIDIA driver 570.211.01, CUDA 12.8.
Workspace	Approx. 900 mm × 800 mm × 640 mm handover table; subject stands opposite the robot; artificial illumination and white-cloth workspace.

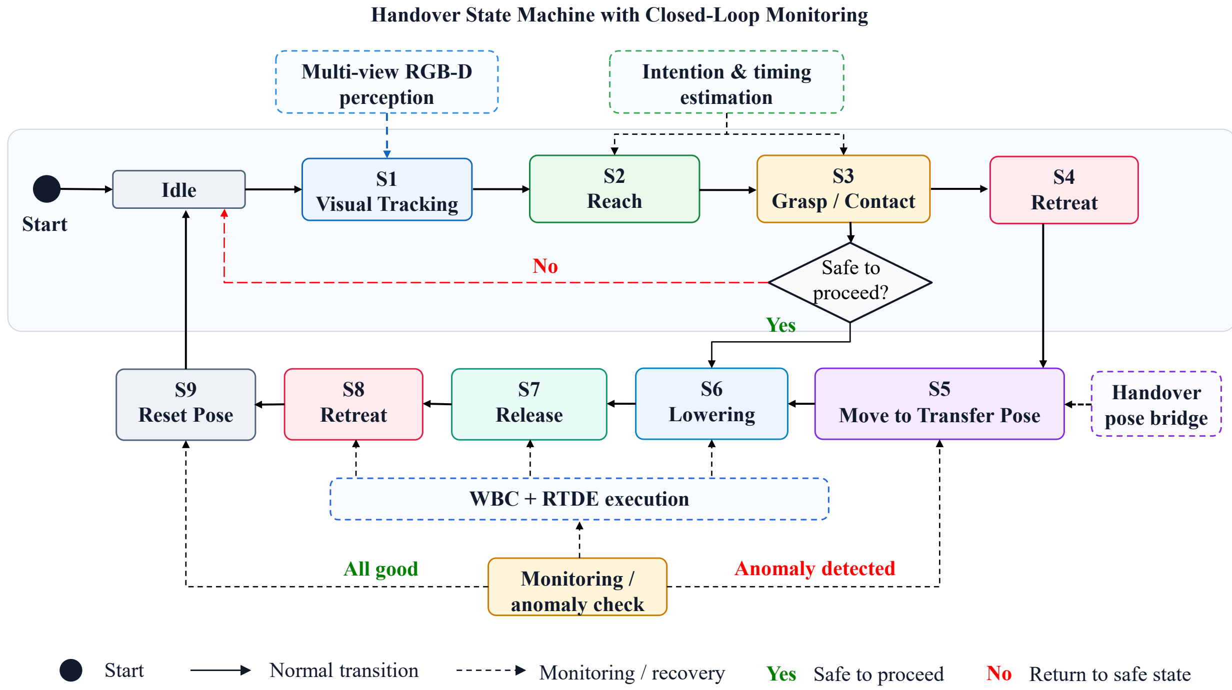

Fig. 2. Robot-side state machine. The controller waits for human motion, follows only fresh and stable targets, advances by a bounded handover offset, closes the compliant gripper, delivers the object, releases it, retracts, and resets.

TABLE II
MAIN SOFTWARE MODULES AND INTERFACES

Module	Function and interface
RGB-D acquisition	Opens both D435i devices directly, aligns depth to color, time-stamps observations, and applies camera-to-base calibration.
Visual parsing	Runs YOLO segmentation for container masks and MediaPipe Hands for one visible hand; outputs cup, hand, left-eye, and right-eye diagnostic points.
Target fusion	Applies semantic, depth, freshness, workspace, class-lock, binocular-disagreement, hand-anchor, and final step-limit gates before publishing the selected target.
Pose bridge	Converts the selected point to `/robot_wrist_pos` and fixed-orientation `/external_target_pose`; invalid or stale targets are not forwarded as new servo commands.
Task scheduler	Implements the handover state machine, target stability checks, interpolation, speed caps, workspace clamps, and recovery/reset transitions.
UR5 driver	Streams RTDE `servoL` targets at 50 Hz with bounded updates instead of executing a single long open-loop trajectory.
Jaw bridge	Translates ROS 2 gripper commands to the local UDP/CAN backend and republishes opening, speed, motor power, command target, and mode telemetry.

The trained visual models are used only to produce candidate masks and semantic confidence. They do not directly command the robot. A candidate must still pass geometric and temporal checks before it becomes a servo target. This separation is important in handover because the visible mask can change abruptly when the human grasps the container from the top, bottom, or side; the control layer therefore treats detector confidence as one input to a reliability decision rather than as a complete pose estimate.

For each camera i, the node forms a container candidate $\mathbf{y}_{i,t}$ with semantic confidence $s_{i,t}$, age $a_{i,t}$, and depth-support count. A hand proxy mask is subtracted from the object mask before depth projection,

$$M_t^{i,\text{safe}} = M_t^{i,\text{obj}} \wedge \neg D(M_t^{i,\text{hand}}), \tag{2}$$

which reduces centroid drift when the human hand covers the cup rim or side wall. Five fingertip landmarks are triangulated from the two cameras and filtered into a stereo hand reference $\mathbf{h}_t^{ref}$. The hand reference is used as a plausibility anchor, not as the object target.

Fresh observations enter fusion only after semantic, freshness, depth, and workspace gates. Important deployed thresholds are: confidence ≥ 0.12, object freshness ≤ 0.35 s, hand-anchor freshness ≤ 0.25 s, at least 8 valid depth points, recognition height ≤ 0.55 m, direct binocular consistency band 25 mm, high-disagreement band 60 mm, track gate 90 mm, anchor-bound publication 8 mm, final 18/35 mm dynamic gates, 6 mm step limit, smoothing 0.70, and 0.8 s reset after missing data.

When both cameras agree, the system fuses the two object centers. When the cameras disagree, it prefers the view that is more consistent with the recent track and the hand-referenced anchor. When no current object observation is legal, the last published point can be reused only inside the same freshness and workspace limits; otherwise the stream becomes invalid. This policy prevents the robot from chasing old positions after the human changes speed or withdraws the cup.

The resulting robust target estimate is a weighted selected/fused observation:

$$\tilde{\mathbf{y}}_t = \frac{\sum_{i \in I_t^\star} w_{i,t} \mathbf{y}_{i,t}}{\sum_{i \in I_t^\star} w_{i,t}}, \quad w_{i,t} \propto s_{i,t}^2 \rho_{\text{track}} \rho_{\text{jump}} \rho_{\text{anchor}}. \tag{3}$$

The final publication stage clips residuals relative to the hand-object anchor and rejects stale holds. This is why the stream is conservative: it may sacrifice some instantaneous localization accuracy to avoid commanding the robot toward stale or discontinuous targets.

III. Robot Control and Gripper

The robot controller consumes the filtered target as short Cartesian updates rather than a long preplanned trajectory. The UR5 `servoL` loop runs at 50 Hz with `lookahead_time`=0.18 s, gain 100, and a target-position update cap of 0.3 m/s. During target following, the scheduler limits target speed to 0.15 m/s and enforces a minimum TCP height of 0.16 m. Runtime target clamps are applied in both the scheduler and stream layer: [0.15, -0.95, 0.10] m to [1.00, 0.95, 0.80] m.

The automatic handover sequence is shown in Fig. 2. S1 moves to an air-wall pose near the cup. S2 follows the fresh target until it is stable. S3 advances 0.12 m toward the container. S4 closes the gripper. S5–S7 move above the delivery point, lower, and open. S8 retracts by 0.18 m and S9 resets. Fixed stages use dense interpolation, speed caps, and slowdown near corners.

The scheduler does not attempt to solve global manipulation planning online. It uses a small number of task states with explicit entry and exit conditions: target freshness, target stability, stage timeout, gripper command completion, and safe workspace membership. This made the behavior predictable during qualification runs. If visual evidence disappears during tracking, the robot holds or slows instead of extrapolating aggressively; if the target reappears inside the allowed region, the controller resumes short-step following.

The gripper backend maps millimetre commands to calibrated motor angle. Homing drives the mechanism until a stall-like condition is detected from low speed and high motor power, then stores the closed reference. The ROS 2 bridge talks to the CAN backend through local UDP command and telemetry channels and republishes `/jaw/state` at 50 Hz. The staged grasp opens to 95 mm before approach, closes toward the object, detects contact from motor power/speed/opening, performs a short inward probe, and then holds or clamps within bounded power. This is a motor-feedback contact reflex, not a tactile classifier.

The tactile assumption is therefore mechanical and procedural. The fingers provide compliance and a broad contact patch; software uses motor power as a coarse contact proxy and stops on simple conditions such as target reach, power threshold, low-speed contact, timeout, or completion of the probe. This avoids claiming physical force estimation, but it gives the system enough feedback to avoid blindly overdriving the jaws during container capture.

TABLE III
Representative validation results

Metric	Result
Target-stream replay	17/17 logs, 66,566 valid events, 96.52 Hz, 92.89% valid output, 140/205 ms median/P95 delay.
Stream quality	7.74 mm RMS, 4.07 mm P95 jump, 10 jumps >20 mm, 4.31% residual energy above 2 Hz.
Component effects	Removing final filtering raises large jumps from 10 to 26; removing reference-guided depth raises RMS to 9.03 mm; raw fused center gives 119 large jumps; removing freshness creates 6.42% stale-held frames.
Task execution	288/288 standardized configurations submitted; task score 95.15; s10=99.97, s11=80.18, s12=100.00, s13=98.74.
Live trials	21/24 valid handovers; frozen-food-bag subset 3/3; median successful placement error 10.00 mm; final mass loss on successful trials 0 g.

IV. Validation and Assumptions

The replay metrics are interpreted as target-stream requirements rather than detector-only accuracy. Valid ratio measures availability, target age and stale-held frames measure freshness, jump counts and P95 displacement measure stability, and output rate plus delay describe responsiveness. This distinction matters because a locally accurate but intermittent estimate can still be unsafe for servoing if it disappears or reappears with a large cross-view displacement.

The ablation results are therefore a failure-mode analysis. The hand cue is not treated as a replacement target and is not expected to improve every valid-frame localization metric. Its role is to provide a plausibility prior when semantic object evidence is partially occluded or when the two cameras disagree. It becomes useful only when combined with semantic container observations, freshness checks, binocular reliability selection, and final publication filtering. Likewise, freshness handling deliberately rejects stale holds even when doing so lowers nominal coverage, because a stale target can command the robot toward a container position that the human has already moved away from.

The 288-configuration task set was balanced over four container labels, empty and filled conditions, three grasp types, three handover locations, and four subjects. This spread was used to exercise the main failure cases: transparent or narrow containers, top grasps that hide the rim, bottom grasps that hide the lower side wall, and lateral handover positions that favor one camera over the other. The live trials were treated as end-to-end checks of the complete perception-control-gripper loop rather than as detector-only tests.

Several assumptions are explicit. First, the system assumes the competition-style layout: a subject opposite the UR5, hand/arm/container visible to the two cameras, and a reachable tabletop workspace. Second, the reported system uses approved local containers; the model filenames and results correspond to this setup. Third, the implementation is vision-first: there is no wrist force-torque sensor, no tactile sensor array, and no learned tactile classifier. Grasp contact is inferred only from gripper motor telemetry and compliance. Fourth, target-stream metrics are replay and execution evidence for this handover domain, not a universal pose-estimation claim.

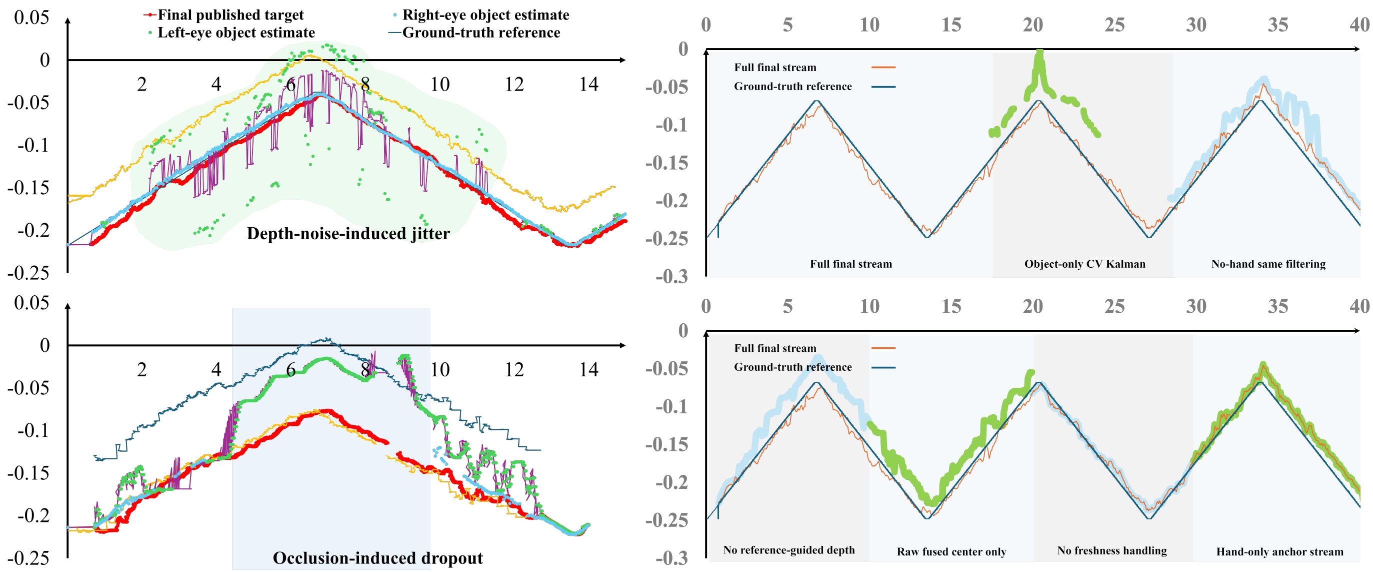

Fig. 3. Target-stream and ablation diagnostics. Left: the final stream follows the target while suppressing unreliable raw or weakly fused outputs. Right: reference-guided depth, freshness handling, hand-reference coupling, and final filtering jointly reduce discontinuities.

V. Design Rationale

The main design choice is to treat perception as a servo input generator rather than as a stand-alone pose-estimation problem. The output must be available for most of the receiving interval, fresh enough to represent the current human motion, stable enough to avoid sudden Cartesian commands, and responsive enough for a 50 Hz robot loop. These four requirements explain why the system uses deterministic gates after the trained segmentation model. A high-confidence cup mask can still be rejected if its depth support is weak, its timestamp is old, its 3D point leaves the workspace, or it disagrees too strongly with the recent track.

The hand cue is used with the same conservative logic. It is not a substitute for the cup target, because a hand-only stream can be biased relative to the container. Its useful role is as a plausibility anchor during hand-object occlusion: fingertip geometry and offset memory make one camera's sudden depth jump less likely to dominate the final command. When the hand anchor is absent or stale, the corresponding reliability factor is disabled rather than hallucinated. This keeps the estimator causal and reproducible in replay.

The gripper follows the same engineering philosophy. Since there is no tactile array or force-torque wrist sensor, the system does not classify contact material or estimate physical force. Instead, it combines passive compliance, low robot speed near the handover, and a simple motor-power reflex. The result is a bounded receiving behavior whose main safety comes from conservative visual target publication, staged motion, and mechanical tolerance.

VI. Conclusion

HARTS is a complete engineering solution for RGB-D-only container handover. The main technical contribution is the control-ready hand-referenced target stream: semantic cup observations are regularized by stereo fingertip geometry, per-view reliability, freshness gates, workspace limits, anchor-bounded publication, and final temporal filtering. This stream feeds a bounded UR5 state machine and a compliant motor-feedback gripper. The result is a deterministic, reproducible handover pipeline with high target availability, low severe-jump frequency, and validated end-to-end execution.